Moving toward coordinated management of timber and other resource uses in Hungarian forests

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This paper presents the relationships and development of forestry and wildlife management in Hungary during the past 70 years. Current problems of coordination are rooted in the cardinal principle of Hungarian forestry that defines the purpose of forest management as "sustainable and increasing wood production", and regards recreational, educational, and wildlife management as secondary uses. Forestry and game management share several common elements such as educational and scientific underpinnings, direction and regulation, and planning systems. Nonetheless, poor coordination and profit-driven management have resulted in serious resource conflicts associated with extensive clearcutting, habitat destruction, diminished diversity in tree plantations, game damage to reforestation efforts, and other problems. Scientific and professional cooperation are necessary to create an ecologically-sound basis to coordinate resource uses and to harmonize the objectives of timber production, nature protection, and wildlife management in Hungarian forests.

Cet exposé présente les relations et le développement de la foresterie et de l'aménagement faunique en Hongrie au cours des 70 dernières années. Les problèmes actuels de coordination prennent racines dans le principe cardinal de la foresterie hongroise qui définit l'objectif de l'aménagement forestier comme étant "la production durable et sans cesse grandissante de matière ligneuse", tout en considérant les aspects récréatifs, éducationnels et fauniques comme étant de second ordre. La foresterie et l'aménagement faunique partagent quelques éléments communs tels les objectifs éducationnels et scientifiques, la gestion et les législations ainsi que les systèmes de planification. Néanmoins, le faible niveau de coordination ainsi qu'une gestion orientée vers le profit ont engendré de sérieux conflits au niveau des ressources, qui découlent de la coupe à blanc sur de grandes superficies, de la destruction des habitats, de la diminution de la diversité dans les plantations forestières, des dégâts causés par les animaux dans les projets de reboisement, et d'autres problèmes. La coopération scientifique et professionnelle est nécessaire afin de créer une base écologiquement éprouvée à partir de laquelle il sera possible de coordonner l'utilisation des ressources et d'harmoniser les objectifs de production de matière ligneuse, de protection de la nature et d'aménagement faunique dans les forêts hongroises.

Introduction

In chronicles, the Carpathian Basin was always mentioned as an area abundant in game. When Hungarians settled in 896, the landscape was quite different; 40–50% of the current territory of Hungary was woodland. The decline of forests started at the 13th century as much was converted into agricultural lands or used for animal husbandry. The peak of deforestation was in the 19th century and it continued until the 1920s (Csöre 1971, Koloszár 1992, Keresztesi 1991).

After World War I, as a consequence of the inequitable peace treaty of Trianon (1920), Hungary lost 67.2% in area, 57.8% in population, and 83.8% in forests. These losses shaped the objectives, principles and development of forestry and game management fundamentally. This paper summarizes the determining elements of the development of use of the two natural resources and their relationships are also demonstrated.

The Period Between the Two World Wars

After the Trianon peace treaty, 10,907 km^2 (10.7%) of the country was forested (Figure 1). Before the war, these forests served only local timber-growing, fuel-wood production, and hunting interests. Conifers occupied 24% of the forests of prewar Hungary, but only 4.1% in 1920. Consequently, the entire softwood demand had to be imported (Keresztesi 1991). The shortage of valuable timber, and unfavourable economic conditions, led to the formation of a forest development concept and an Act (1923) including the afforestation of the Great Plain, the introduction of natural regeneration methods, and the extension of planned management to all forests of the country. The purpose of these efforts was to achieve self-supporting timber production, reduce the costs of import, and reproduce the national forest resources on an increasing scale (Oroszi 1992). The new Forest Law (Act IV 1935) reinforced these aims, suspended free circulation of private forests and made planned management compulsory in all forests. In spite of legal efforts, the area of forests remained about the same between 1920 and 1945 (Figure 1). The reasons for failure were that planned management seriously conflicted with the profit-interests of forest owners, banks did not give credit for afforestation, the state could not subsidize afforestation considerably, small land-owners had no professional knowledge, and the administration was insufficient for the task (Oroszi 1992, Rakonczai 1993).

Between the two wars, game management and hunting were regulated by the Hunting Law (Act XX 1883). Hunting rights belonged to the land and the land-owner could hunt on his property if it was greater than 200 cadastral acres (114 ha) in one block, or the hunting rights on a given area were leased out by municipalities. The law made the land-owners or the lessees (to whom the hunting rights belonged) responsible for the agricultural and forest damage caused by red deer, fallow deer, and wild boar (Benkó 1935).

Hungary was regarded as the European country most plentiful in game, attributed to the interactions of fertile soils, mild climate, and a rich flora. The general abundance of wildlife was accompanied by the excellent trophy quality of deer and at the World Hunting Exhibition held in Berlin (1937), Hungarian red, fallow, and roe deer trophies won many medals and a majority of grand prizes (Kiss *et al.* 1942).

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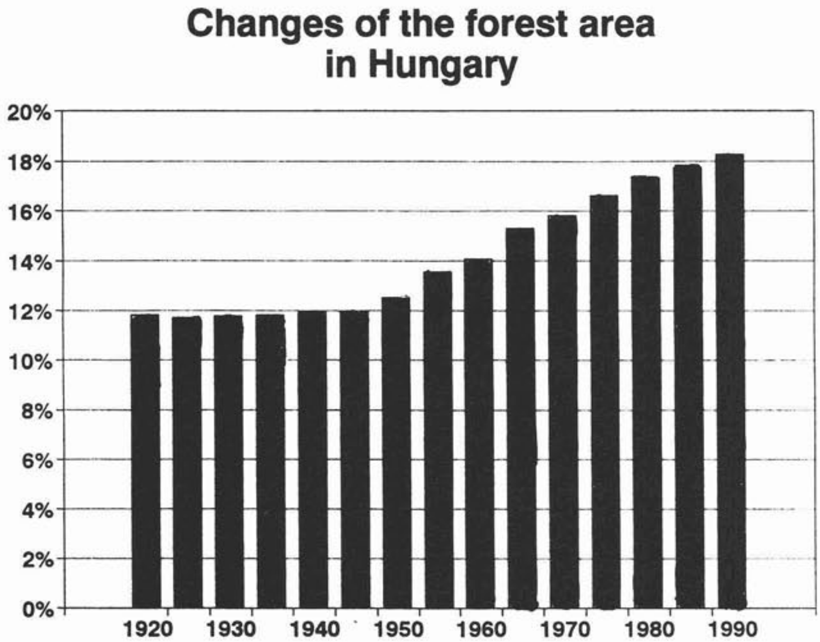

Figure 1. Percentages of forested area in Hungary, 1920-present.

Managers maintained red deer, roe deer, and wild boar in large numbers only on private properties, where serious forest and/or agricultural damage could remain private matters (Benkö 1935). Generally, agricultural damage from wildlife was considered a major problem, and forest damage was secondary or overlooked (Holdampf 1962). To improve deer-trophies, selective shooting was advised and, knowing the negative effects of over-crowding, strong hunting pressure was recommended. This big game management for quality was also in accordance with the aim of harmonized relations between game management and forestry/agriculture (Benkö 1935, Kiss *et al.* 1942). In order to reduce game damage conflicts, a system of hunting districts was proposed to manage big game populations under the same principles and to minimize the debates about the origin of damaging animals (NVV [National Society for the Promotion of Hunting] 1940).

On large estates, the balance between the interests of forestry and game management was mostly determined by the landowners, and hunting could be an important source of income. Hunting tourism and game trading was regarded as a competitive source of hard currencies on the national level (Kiss *et al.* 1942). That was why the *res nullius* (ownerless) status of game was debated and to consider it national property as well as state-ownership of game were suggested (Roth 1940).

It was accepted that forestry and game management are analogous and their development shows parallel elements; "afforestation and replantation" have the same role in forest management as "breeding and rearing" in game management (Benkö 1935). Indeed, foresters were expected to be educated in game management. Game management was a compulsory subject in the curricula of specialized secondary schools and college forestry (Bencze 1971).

Development of Forestry After World War II

The war still raged in Hungary when the provisional national government passed the Agrarian Land Reform Bill to abolish the system of *latifundia* and to make a re-allotment of land to landless peasants. In order to meet the public demand and to realize modern, intensive forestry, the majority of large forest estates became state property. All state and joint-tenant properties (89.1% of forests) were managed by 15 national enterprises, permitting the stepwise introduction of long-term forest management principles. The main policy goals were the reproduction of growing stands on an increasing scale, multiple use of forest resources, mechanization (industrialization) in forestry, and reconstruction and development of the timber industry on the basis of domestic raw materials (Keresztesi 1991).

Therefore, the Great Plain afforestation plan was broadened into a national program, and between 1946 and 1985 about half of the abandoned or non-profitable agricultural land was afforested (Fig. 1). Accordingly, forestry and the wood industries developed very dynamically and the attitude of foresters changed markedly into a technical-economic approach, including plantation, harvesting and rational use of timber. As a result of this dynamic growth, more initial plantations (natural regeneration or reforestation after felling and afforestation) were established than the sum of all forested area after the war; large-scale afforestation facilitated the rational use of marginal lands unsuitable for agricultural production; coniferous and soft broad-leaved industrial wood-producing plantations were established and existing forests were converted to the desired types. This resulted in a 2.5-fold increase in the growing stock and the felling potential. The mechanization and efficiency of labour increased dramatically. The timber supply/demand situation of the country improved and timber import declined (Halász 1991).

About 70% of the Hungarian forests are of human origin. The mechanization of forestry fundamentally changed the methods and extent of all processes and forestry became, in many senses, similar to agriculture. Compared to natural silviculture (natural or close-to natural self-regulating forest ecosystems), silviculture of man-made forests means principally production of the highest volume of timber in the demanded species in the shortest time, considering other functions (protection, recreation) of forests. Hungary is in the central European deciduous forest zone and the timber demand of the economy determined the species composition of new forests. According to the principles of species-policy on all sites, target stands (tree species) have to be established that can use the potential of the site the best way and yield the highest value per hectare (Járó 1991). Consequently, the ratio of oak forests increased and fast growing poplar species, black locust, and conifers were planted in all areas where site conditions, silvicultural, and economical reasons justified their introduction and cultivation (Keresztesi 1991).

In 1961, the Forestry and Game Management Act (Act VII 1961) founded the long-term framework of forestry and game management. At the same time the Law on Land-Use, and the forced establishment of agricultural cooperatives, rearranged the ownership relations and about 30% of forests became managed by large farms. The economic reform of 1968 did not help the use of natural regeneration and the annual profit-orientation came into conflict with the long rotation cycle of forest management. The failure resulted in concentrated felling in high quality stands, delays in reforestations, large clear-cuts, and increasing cost-sensitivity (Keresztesi 1991).

The pressure on forest enterprises to achieve short-term cost-effectiveness, as well as profit-oriented management, resulted in large forest complexes with low diversity of tree-species and uniform age distribution that provide insufficient food supply and inadequate shelter for game (Köhalmy 1992). The rough and ecologically-intolerable ways of forest management have generated strong criticism urging an increased role for native species, exile of exotic species, more diversity, and forest wilderness. Accordingly, in the current forestry literature, more attention is paid to the preliminary conditions and an ecological basis of close-to-natural forest management methods. These methods include appropriate selection of tree-species, conscious combination of species, increased number of seedlings planted per hectare and protection against game damage (Koloszár 1992, Führer 1993). On the other hand, to reduce the contradictions, more attention should be given to the habitat requirements of game and more forests could be designated with game management as the primary use (Tóth 1987).

Development of Game Management and Hunting After World War II

In 1945, the government proclaimed game as state property and separated the hunting rights from landed property. The Ministry of Agriculture defined some 2000 hunting territories and leased them to the hunters' societies or gave them to the state forest enterprises for management (Tóth 1991). Membership in a hunting society is a prerequisite to apply for a gun and to hunt. All members must participate in works of the society on a volunteer basis providing inexpensive dilettante labour.

The main task of game managers was to re-establish game populations and set the basis for future management. Particularly deer were protected and the improvement of impoverished big game stocks was considered a responsibility of state forest enterprises (Szederjei 1956). Licensed hunting for foreign hunters was restarted and these revenues were the basis for proposals suggesting the development of game management and hunting, as well as a 50% increase of hard currency incomes prescribed by forestry authorities (Holdampf 1962, Tóth 1964).

By decree of the Council of Ministers in 1954, big game management in forests owned or managed by the state was put under the direct control of the General Directorate for Forestry. Although big game management was clearly connected to forests and coordination of game and forest management was of primary concern, the problems of forest damages were discussed (Bencze 1960, Balsay 1962). It was a widely-held opinion that, since the beginning of the 1950s, big game populations (mainly red deer) were much above the carrying capacity of the forests and should be reduced (Keresztesi 1991). Damage seemed to be intolerable in conifer afforestations, poplar plantations, extensive reforestations, and oak and beech regrowth. In spite of these contradictions, forest area and allowable timber cut increased rapidly (Tóth 1991).

Between 1958 and 1961, the introduction of a uniform long-term planning system started in order to harmonize game management with the requirements of national economy, forestry and agriculture. The General Directorate of Forestry was responsible for development of game management and uniform direction, including planning. In the planning system the meaning and practical use of carrying capacity were argued. According to the official decision, forest carrying capacity and game damage were directly correlated, and the number of big game which could be supported on a given area could be determined on the basis of the available food. This approach did not consider other needs of game, such as shelter, water, and freedom from disturbance, and was demonstrated as a misconception (Balsay 1962). The biological (food) carrying capacity of forests is different from their economic carrying capacity or game-tolerance (Balsay *et al.* 1972).

The Forestry and Game Management Act (Act VII 1961) integrated forestry and game management into the centrally planned economy. It determined the most important task of game management: to develop game stocks by rearing and protection. The Head of the National Forest Authority was responsible for the uniform direction of game management. In 1968, the direction of forestry, wood industry, and game management was taken over by the Ministry of Agriculture and Food, and hunting administration became the duty of country committees (Tóth 1991).

In order to improve planning it was declared that the data and tasks connected with big game management (game carrying capacity, game feeding plots, etc.) should also be prescribed in forest management plans (Tóth 1991). Forest Management Planning Service prepared the plans and their primary purpose was to adjust the number and species composition of overpopulated big game to the natural game carrying capacity of forests. Based on environmental conditions (circumstances of agriculture and forestry in a given area), the food consumable by game was determined. This method categorizes forest stands into 5 classes (excellent, good, average, poor and very poor) and the carrying capacity is calculated in red deer units, in which 1 deer unit = 2 fallow deer; 3 roe deer; 2.5 mouflon; or 5 wild boar (Balsay *et al.* 1972, Tóth 1991). Plans were prepared for about a third of the hunting territories but the number of big game could not be reduced and their population growth continued (Fig. 2). The insufficiency was mainly explained with the incomplete (mosaic-like) coverage of planned management units (Fatalin 1984).

It is inconsistent that the Forestry and Game Management Act determines the responsibilities and rules for compensation of agricultural damages but does not include similar regulations for forest damage. The increase of game numbers resulted in the rise of agricultural and forest damage, and the latter became a never-ending problem (Fig. 3). Damage that com-

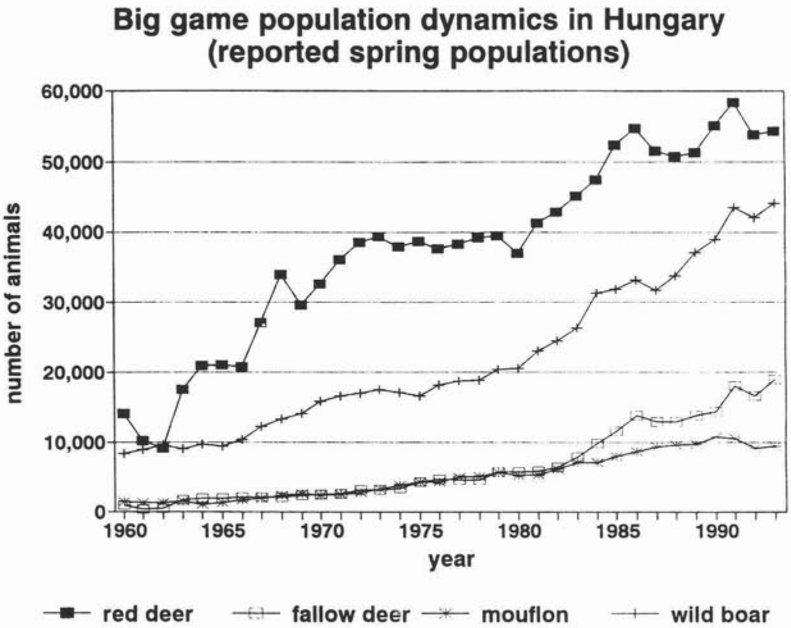

Figure 2. Estimated populations of large mammal species in Hungary, 1960-present.

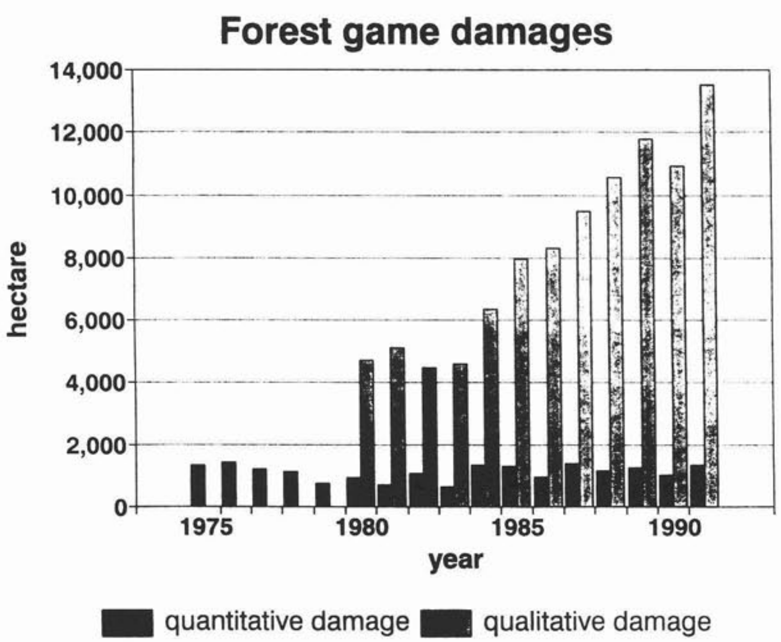

Figure 3. Damage to forest stands from ungulate browsing, 1975-present.

pletely destroys a plantation or afforestation, necessitating replanting, is referred to as quantitative damage. Qualitative damage destroys some portion of saplings or trees, and recovery is possible. In spite of the dynamic increase of big game populations, quantitative damage showed a waving trend. The tendency of qualitative damage was similar to the red deer and wild boar population dynamics. The unsettled situation of forest damage compensation (and compensation for related costs such as fencing) became worse as cost-sensitivity of forest enterprises grew.

In 1981 the official guidelines for game management and hunting were renewed but the objectives did not change: (1) to provide a suitable basis for recreational hunting, (2) to increase the revenues of game management, especially hard currency incomes, and (3) to maintain the trophy quality of big game. To mitigate the tension between forestry and game management, new categories of big game kept in fenced reserves, and big game sustained with supplemental feeding, were introduced. All these efforts were emphasized by the establishment of a unified game management planning system in the whole country. The sustainable game numbers were determined by the Forest Management Planning Service and a population increase was planned for roe deer, fallow deer, and mouflon; a population decrease for red deer and wild boar. The latter two species caused the most damage (Fig. 4) in forests. The role of Forest Inspectorates was to be increased, and changes in the legal and administrative framework of compensation for forest damages were planned (Mátrai 1981, Tóth 1991).

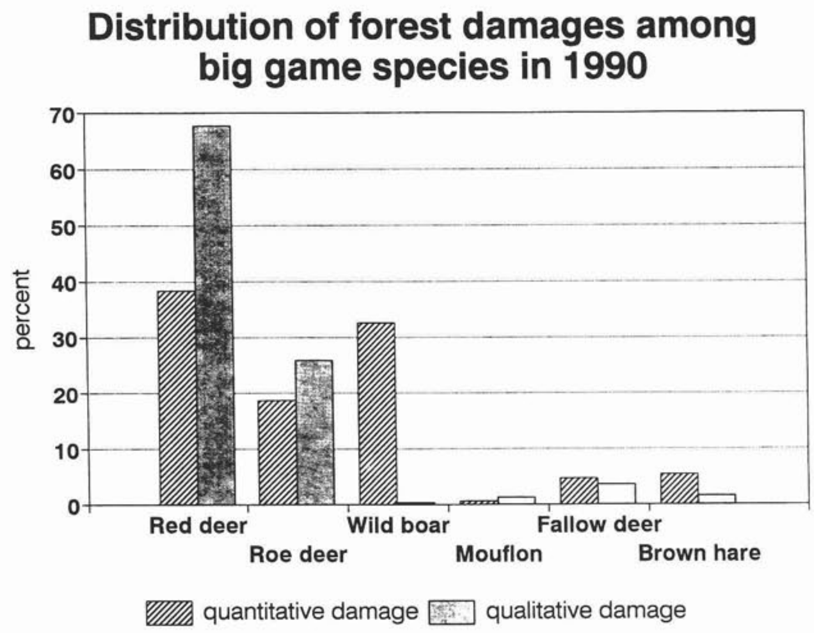

Figure 4. Proportion of forest damage by browsing attributable to the various game species.

As forest game damage remained the main problem in the subsequent years, the Forest Research Institute was entrusted with research of forest-game problems (Bondor *et al.* 1989). Moreover, to diminish the discrepancies, plantation of more diverse game feeding plots, better cooperation of forest enterprises and hunting societies, analysis of the relationship between game density and threshold of forest damages, development of objective forest damage estimation methods were urged (EFH [University of Forestry and Wood Industry, Sopron] 1983, EFH [Office of Forestry and Wood Industry, Ministry of Agriculture and Food] 1987). It was also suggested to designate game management as the primary function in more forests than the actual 35,000 ha (Tóth 1982). Further, the plantation of fodder forests was suggested, which could be established on marginal lands unsuitable for profitable agriculture or could be developed from low-yield forests inappropriate for wood production (Keresztesi 1991).

Out of the significant problems of the planning system, the lack of methods for population estimation and the improper determination of harvests should be noted. These resulted in the serious underestimation of big game populations (especially red deer and wild boar) and an under-harvest, with unbroken population growth (Csányi 1989, 1991). The problems were intensified by the loose connection between the forest and game management plans, and the fact that game management plans could be neglected without consequences (Fatalin 1984).

The efficiency of the planning system was first evaluated in 1986. The strong inconsistency of the global interest of long-term timber production and the profit-interest of state enterprises, hunting societies and game-trading companies were clearly evident. This contradiction was most pronounced in the case of state forest enterprises. There, in order to increase the game stocks, an additional 90% of the deer were supplementally fed. Between 1971 and 1984, game management was the best-paid (25%) branch of state forest enterprises, while the profit rate of felling was 23% and forest regeneration was 14%. This high profit rate motivated several state forest enterprises to obtain hunting rights in as many areas as possible (Mátrai 1990). The economic attractiveness of game-business was also illustrated by agricultural cooperative farms that wanted to take over the hunting rights from hunting societies, reverting to the traditional rights of land-owners or land-users (Tóth 1987).

The need for legal changes was recognized and a new Game Management and Hunting Act was planned, in which the measurement, estimation and compensation of forest damage would be legally settled, and the role and power of local forestry inspectorates and hunting authorities would increase (Tóth 1987, 1991).

Conclusions and the Current Situation

According to some foresters, forests should conform to the following functions: production, protection, and recreation, with timber production as the priority. Game management is one kind of supplementary gain from forests. Sustainable management in forestry aims at continuous production and use of resources without any adverse effect or decline (Bondor *et al.* 1989). It is recognized that public expectations and claims are changing and more complex social and ecological questions should be addressed. According to the concept of welfare forestry, forest management is concerned not only with the sustained and economical production of maximum quantity and best quality timber and other products (fruits, honey, mushrooms, medical plants) but also with the intangible benefits of forests (quietness, recreation, hunting, landscape attributes, nature protection). That means the multiple use of forests, but recreational and other functions of forests should be enforced without jeopardizing their wood-producing basis (Keresztesi 1991). Except for game management, the changing attitude is somehow expressed in the distribution of forests according to their primary function (Table 1).

Forestry and game management developed together during the last decades, especially after World War II. Although both developed significantly, serious resource conflicts appeared during the last 25 years. The reasons for this failure and the necessary changes/efforts can be summarized, as follows:

Table 1. Changes in the primary functions of forests in Hungary.

	1965	1980	1985	1991
		(%)		
Wood production	94.1	81.2	80.3	79.0
Special functions	5.9	18.8	19.7	21.0
Recreation	0.5	3.5	3.5	4.8
Protection and nature conservation	5.1	12.5	13.7	14.0
Hunting and other	0.3	2.8	2.5	2.2

• The long-term interests of sustainable forestry have become contradictory with the profit-driven management of the state enterprises. A similar discrepancy appeared in the case of hunters' societies for which more big game offered more shooting opportunities for individual hunters, as well as their contribution to the national hard currency income was demanded. In a profit-driven environment the behaviour of enterprises is largely determined by their short-term interests.

• Although game management plans could be ignored without consequence, these problems cannot be solved only with administrative tools. If the control system does not change and the feed-back is not strengthened the problems will remain the same.

• The carrying capacity of a given area cannot be determined on the basis of the available food because it is greatly influenced by environmental factors that also affect damage by game. While both foresters and game managers are talking about natural game carrying capacity, the real aim is to keep big game populations at minimal or no impact densities.

• The lack of reliable population estimates makes harvest plans powerless, vulnerable and debated. The assumptions of the planning system are static and deterministic. The practical connection between forest plans and game management plans is minimal and not informative. A more dynamic system is required based on valid data and able to follow year-to-year population changes.

• Fencing of plantations, as the only effective solution against damages, created a vicious circle: damage — fencing — habitat/food loss — more damage — new fences — further habitat loss — and so on.

• In the future, more attention is needed to improve game tolerance of forests and afforestations, as well as to develop and apply more environment-friendly protection methods. It is necessary to enhance our knowledge about the space- and time-use by big game species to understand better their behaviour and roles in ecosystems. The determination of game management regions supports the concept that in ecologically similar areas, identical game management strategies must be applied for state enterprises and hunters societies.

The political changes in 1990 brought new opportunities, expectations and dreams, as well as disappointments, frustrations, and failures. Unfortunately, solutions are expected from politicians, namely new Forestry and Game Management Acts, or perhaps from a new Act combining Forestry and Game Management.

Several changes and events illustrate the uncertainties. The lands of agricultural cooperative farms have been privatized and one third or more of the forests will be private property. These forests can be over-cut, not regenerated according to the standards, and the pre-war situation of forests will return. According to the official standpoint in forestry, the decrease of profit-orientation is desirable, and efforts should be concentrated on the increase of quantitative and qualitative production of material products and the intangible benefits of forests. Forest area cannot decline and the new forestry policy must be focused on protection as the main function of future forests (EFF 1993). In fact, the 19 state forest enterprises are under the control of State Property Management Holding. This organization manages all state enterprises and is a profit-driven corporation.

In order to use the large areas unsuitable for rational agricultural production, further enormous afforestations are planned. The ecological basis, species-selection, potential methods, expected results and predicted success, as well as the needed and available financial basis are strongly debated or not clearly foreseen (Rakonczai 1993).

It is the latest political decision that hunting rights should belong to the landed property. This way land-owners who control game management and game damages, can also get their share from the profit of hunting or the existence of game. In order to reduce the forest game damages, red deer, fallow deer, and mouflon shootings were increased and a population decline started. The future of hunters societies has become uncertain, several legal constraints are violated, and poaching is a serious problem throughout the country.

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